

Shanrong Wu

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EDUCATION

University of Illinois Urbana-Champaign

Urbana, Illinois

Bachelor of Science in Computer Science — GPA: 3.94 / 4.00

Expected Graduation Date: May 2029

Relevant Coursework: Data Structures, System Programming, Database Systems, Computer Architecture, Software Design Lab, Open Source Software for Education

TECHNICAL SKILLS

Languages: C++, Java, Python, Rust, LabVIEW, MIPS Assembly, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks: React, Next.js, TailwindCSS, ROS2

Developer Tools: Linux, Git, GitHub, GitLab, Docker, VS Code, NeoVim, Supabase, PostgreSQL, Shell, NeoVim

Other Tools: Microsoft Office, Excel, Word, and PowerPoint

PROFESSIONAL EXPERIENCE

SIGrobotics - F1tenth

September 2025 – May 2026

Simulation/Planning Group

Urbana, Illinois

- Develop autonomous driving functionality for the F1TENTH racecar in the F1TENTH Gym simulation using C++, ROS 2, PID control, and path planning/control algorithms across multiple track layouts
- Support development for the physical racecar by working with 2D LiDAR, stereo vision, and Jetson Nano
- Engineered a local planner in Python and ROS2 Humble that integrated global planner output with robot pose data to generate waypoints for pure pursuit based navigation

SIGmobile

Oct 2025 – May 2026

Backend Developer

Urbana, Illinois

- Developed backend for CS Course Recommender, a mobile app designed to recommend CS courses to students
- Collected and processed course data by scraping UIUC Course API and assigning interest tags to each course
- Wrote Python scripts to fetch and parse various course information from an UIUC Course API into a JSON

FIRST Robotics Competition

August 2021 – May 2025

Software Lead, Technician, Programmer, Electrical

Bradley, Illinois

- Mentored 20+ team members in Java and LabVIEW with hands on demonstrations and programming activities
- Fixed electrical and software issues under competition time restraint and pressure to maintain robot reliability, reducing match time failures by 5+ matches per season and leading to an increase in overall rankings
- Led the migration from LabVIEW to Java and upgraded software and hardware systems for long term support

PROJECTS

HackIllinois 2026 – HackAstra | React, Python, Flask, Gemini API, Supermemory

February 2026

- Led a team of 3 to develop a full stack AI inspection software application for the Caterpillar Track, automating daily inspections through audio, video, and image-based input, using React and Flask
- Integrated the Gemini API to deliver real-time AI generated inspection report during the inspection workflow
- Implemented an inspection history to store and display previous inspection data and combine same day reports

RusTX - Raytracer in Rust | Rust

October 2025 – January 2026

- Developed a ray tracing application in Rust with a team of 3 to render a scene using STL files and randomization
- Designed and implemented geometry processing utilities to convert 3D coordinates and STL mesh data into triangle objects for ray object intersection testing using the Möller-Trumbore algorithm
- Optimized runtime by about 10x with parallel processing, multithreading, and denoising techniques

IlliniBites | TypeScript, React Native, Expressjs, HTML/CSS

Sep 2025 – Dec 2025

- Developed a mobile app called IlliniBites with a team using React Native framework
- Improved user experience by implementing filtering and sorting feature on specific topics
- Designed and implemented a backend API with Express.js for efficient retrieval and delivery of restaurant data

Paradox | JavaScript, React, Next.js, HTML/CSS, TailwindCSS, Supabase, SQL

July 2025 – August 2025

- Developed an interactive web application using Next.js to track and manage various data for the Paradox clan
- Improved data logging efficiency by upgrading from manual Google Sheets workflows to a Supabase database
- Designed responsive interface to display user and clan metrics using HTML/CSS and Tailwind CSS